Bias in transfer prices of prime icons in EA FC Ultimate Team?

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Abstract

Soccer or association football transfer prices for players at the moment are generally seen to be at their highest and at their most volatile point ever, to the point many experts believe it will begin affecting the integrity of the game. Many factors are taken into the valuation of a player's value and many of these factors may have no relation to the player's ability to play soccer but rather volatility and instability in transfer prices are the transfers of Neymar da Silva Santos Junior commonly known simply as Neymar and Harry Maguire the English center back. Neymar is currently the most expensive transfer ever in soccer, he cost Paris Saint Germain approximately $245 million to bring him to Paris from Barcelona. This price tag is before his wages are taken into account which cost Paris Saint Germain about $42 million each year. Neymar is one of the best players in the world but many questioned the obscene fees paid for his transfer which shaped the market since. Some conspiracy theorists speculate that PSG's ability to pay these amounts was because they are indirectly owned by the Qatari government, who wanted to use Neymar to help them promote

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their World Cup hosting bid. Usually such transfer clauses were previously put in place as a safeguard to keep the player in the club. — December 2019

As for the case of Harry Maguire, one must look at the infamous slip of Steven Gerrard against Chelsea in the 2013–2014 season that some say caused Liverpool to lose the title. Are his errors on the pitch freudian slips, or is it his way of communicating to his team where they are losing the match. For a player to have that price tag and make as many alleged errors as he does, one must try to understand the meaning behind the price tag. Is there such a thing such as a tactical error for a defender ? — November 2024





We are able to gain a clearer view of the complex interactions involved in soccer transfers from the following passage obtained from Nobel Prize-winning economists Abhijit V. Banerjee and Esther Duflo's book, *Good Economics for Hard Times*.

" *One thing the absence of caps in European soccer and the resulting*

astronomical salaries does is encourage players to evade taxes. In 2016, Lionel Messi (who made more than €100 million in 2017) was found guilty on three counts of defrauding tax authorities of €4.1 million and given a suspended jail sentence. In July 2018, the Spanish government and Cristiano Ronaldo signed a deal in which he agreed to pay a fine of €19 million and receive a suspended ***prison*** *sentence. He was accused of four counts of tax fraud worth €14.7 million, resulting from the use of shell companies outside Spain to hide income made from image rights from 2011 to 2014. Moreover, many of those who do not actually cheat shop around for lower taxes. Comparing countries in Europe that raised or lowered taxes at different points in time, a study found that when the tax rate in a country increases by 10 percent, the number of foreign players goes down by 10 percent. In 2018, Ronaldo left Spain for Italy to lower his tax bill*" — Excerpt From *Good Economics for Hard Times,* Abhijit V. Banerjee and Esther Duflo

Can the prices of players in the EA FC online market provide a different perspective in judging the value and capability of players ?

So what is Ultimate Team?

Ultimate Team is an online market where players can trade cards, that represent players in the game for EA FC UT coins. Players can trade current players, and legends in soccer in this feature. The cards highlight the main qualities of the players, together with an overall rating of how good the player is. EA Sports puts in a lot of work to rate their players as accurately as possible and goes into great detail, even including cognitive attributes such as **vision.**

A glimpse of Mesut Ozil's impeccable vision, on the left are **Phoeniconaias minor** in a lake

Upon looking at the rating system, you would expect that players with higher overall ratings to have a higher price, and players with similar qualities and attributes overall to have similar prices. This however is not the case within the game. In fact, Pelé, the highest rated player in the game, is not the most expensive player on the market. Even complete players are not passing the ultimate team market litmus test. Players that

can hold the line or make tackles or tactical fouls when it counts are not particularly of great value in the game. Although Claude Makélélé had a role named after him and revitalized the defensive midfield position, his cost does not reach one million coins (more on Makélélé and Chelsea at the end of the article). This is concerning as some upcoming stars such as Endrick use the game and the icons to learn the playing styles of old legends of the game.

> *There is no doubt that the Serie A was and is the hardest league to play in. The defenders and goalkeepers are simply the best on the planet, and the tactics are a combination of subtlety and brutality. Winning or losing is all-important in Italy, so if an attacker gets past you, he must be brought down. The 'tactical foul' is a way of life for Italian defenders — and not to be confused with the 'useless foul' from which your team takes no advantage. To be top scorer in Italy was truly a formidable task.* Calcio, a history of italian football by John Foot

Although footwear technology has improved together with nutrition and body care, the play-styles of these icons provide great insight into the development and fundamentals of the game. In regards to the most expensive player on EA FC FUT, who is this mystery man that is of value in Ultimate Team, the man with the billion dollar briefcase ? That title goes to Ronaldo, O Fenomeno. (In EA FC 25, Ronaldo, O Fenomeno's card is the most expensive in the market costing 12, 800, 000 coins).



Ronaldo, O Fenomeno

When looking at this fact, we can tell there is more to the price for which the market decides to pay for the player, such as how the individual attributes of the players work with the gameplay mechanics. As someone who played Ultimate Team as a teenager, this interested me enough to want to figure out what made it tick.

Overview of the Problem

The problem presented above is not something new, it is widely accepted that some iterations of Ultimate Team have had some attributes that were overpowered within the game. In earlier iterations of the game, it was not uncommon for less known players in real life to be extremely popular within ultimate team because of having high values for these attributes. Some honorable mentions are Emenike of FIFA 13, Adebayo Akinfenwa of FIFA 13, and Seydou Doumbia of FIFA 15. This issue has been discussed in detail by some, an example is this article discussing the use of pace in FIFA 15. In this article, I will try to show this bias by creating a

hedonic model for the prices of players using physical and technical attributes provided by EA Sports on their players and prices.

In one of the recent iterations, FIFA released a special group of players into Ultimate Team called Icons. Icons are retired soccer players who are considered legends of soccer. Each Icon has 3 cards, Prime which is supposed to reflect the capabilities of the legend in the prime of his career, a medium card and a base card. All the icons also have an overall rating, there are about 122 players designated to have icon cards in the game and their prices range from about 60,000 coins to 13,000,000 coins. Icons are good to study as they are stable in terms of price compared to the cards of current players who can receive special cards with upgrades depending on their form which can easily sway the prices of earlier cards (this was a comment I wrote when I was in a flow state, the proof of this economically/mathematically may take some time).

Among the icons, I will be looking at the impact of having dominant physical traits that reflect strength, speed and dexterity on the prices. I will be trying to see if players that display these physical traits are more valuable in FIFA 19 compared to players whose main quality in soccer is their creativity and in-game IQ and hope to see that these attributes are dominant in the determination of the price of the players. One statistic that highlights the possible bias in physical traits in FUT is that Cristiano Ronaldo's base card was worth 5 times more than Messi's card in previous iterations of the game even though they both have very similar overall ratings.(In FIFA 20, Messi's base card cost more than Ronaldo's base card).

NB : For clarity, goalkeepers have not been included in this study.

The Blind Sayeed

Amongst the features in the game, is the ability of players to have managers in their team. Initially, this may be seen as an opportunity to have EA FC UT players unlock different tactics to use in the game based on the choice of manager. Instead, the purpose of the manager in EA FC UT is to add chemistry points and provide a minute increase in the percentage of a player's contract in the game. More on chemistry points later in the article.

In addition, there is no distinction between a manager and a head coach. They are all treated the same. A head coach has less responsibilities than a manager who also has the ability to direct the purchase and the vending of players. A good example to show the difference is Mikel Arteta and Pep Guardiola. Mikel Arteta is a head coach, so can direct all his energy into improving team morale and working with the players he has. He has a football director to help him with the club's transfer decisions. Pep Guardiola is a manager and has more control in the flow of transfers in and out of his club. A good example is how a player of Cole Palmer's quality was an afterthought in his squad due to his preference of Phil Foden and not due to squad issues.

they said you have to do the homework, I said $#@! [the] *homework put that boxing tape on, I knew what I wanted to do, I wanted to be champion* — Floyd Mayweather Jr

Does the fact that Mikel Arteta doesn't think about transfers as much as Pep Guardiola make him any less of a manager than Pep Guardiola ? They have been the dominant competitors in the Premier League for both 2022–2023 and 2023–2024 seasons.

Introducing an element of risk — The Referee

Plutarchs Life of Lucullus *provided and early, literal example: the king of Armenia got advance notice that Lucullus's troops were approaching. He killed the messenger for delivering that message, henceforth, messengers stopped reporting such intelligence. Obviously, if you don't like the message, you shouldn't take it out on the messenger.*— Thinking in Bets, Annie Duke

In my first contacts with console games as a child, be it with PES or FIFA, I remember that as a "cheat code" you could purposely dive to influence the referee's decision on the status of a player on the opposing team in some of the games. This was a double edged sword, if you did it the

wrong way, you yourself would be given marching orders.

In the current state of play, referees are expected to be nearly perfect in the game leaving no margin of error. If a referee commits an error it is assumed to be a bug. In real life, despite the introduction of VAR and Goal Line Technology to improve the quality of decisions on goals and offsides, many fans still find themselves disagreeing with referees' decisions on such things such as penalties, fouls, yellow cards and sending offs. Each referee has a different temperament, and some referees have sterling reputations such as Mark Clattenburg or Pierluigi Collina.

Its possible that an introduction of similar cards for referees highlighting their tendencies to view things such as sliding tackles repulsive or their leniency for giving a penalty for fouls done in the box would make the game more fun by introducing an element of risk. This could also act as an incentive for players to move up in divisions in rivals or go further in

the weekend league, as they would get the opportunity to play with higher quality referees.

The theoretical probability of chemistry points — 2024 addition

Neymar's Value in FIFA 23 (EA Sports)

Neymar's Value in EA Sports FC 24

We can see within one year the value of Neymar's base card dropped by over $100,000 coins. The only significant thing that changed from his average ratings was a one digit drop in his average pace rating. What could be the reason for such a precipitous drop in the value of his base card ? The answer could lie in EA FC chemistry points.

"What is chemistry exactly, and why is it important? Well, chemistry plays a crucial role in the performance of your players. Each of your players can have from zero to three chemistry points, and the higher the chemistry, the better he will play, as the player will receive bonus stats according to his chemistry style. It is important to note the players will only get bonus stats, so if a player is on zero chemistry points, he won't be negatively impacted." (Source: teamgullit.com).

"When you have your players in the correct positions, you can get them to full chemistry. Players can earn chemistry points by sharing the same

country, league, or club with other players in your team." (Source: teamgullit.com)

As mentioned in the abstract, the nationality of a player can play a big factor in a professional soccer player's transfer fee. The following meme that was circulated after Declan Rice's transfer to Arsenal provides some perspective on this.

The same applies to EA FC UT players but in a different way. Think of a player's club, nationality as factors that influence the number of teams a player can play in. The club has an additional factor it adds upon, the dimensionality of a player which is the league the club is in. We can create a probabilistic theory regarding these two factors involving permutations i.e the number of teams these facets of a player allow them to play in.

Media4Math

Take an English player, just by the fact that he is English, he is very likely to be playing in England. Why? Lets take a look at the starting line up of England in the Euro 2024 Final.

England Starting Lineup vs Spain, Euro 2024 Final (Source : Google)

The only players who played abroad from this starting lineup are Harry Kane(Bayern München) and Jude Bellingham(Real Madrid), and Harry Kane had spent 13 years playing in the English Leagues.

Former England Manager, Garry Southgate attending different soccer matches. Almost all of them are in England. It was his decision on who makes the England National Team Squad.

His game attendance implicitly implies that homegrown players get a preference in selection to the England National Team providing an incentive for English players to stay in England, unless you are playing for Real Madrid...

What is the benefit of this ? Statistics show that the English Premier League is the most watched league in the world. By having the best English players play in England, the name recognition of English players increases and this may be reflected in prices of English players.

So what about Neymar ? He moved from the French league to the Saudi League. It is very likely that this move would drastically reduce the potential number teams he would be able to play in. It says a lot about Neymar that his price in EA Sports FC 24 was still 141, 201 FC Coins after his move to the Saudi League. We can create a sketch of the possible theoretical probabilistic formula on the impact of club and nationality on

the price of the player.

Once quantified, the result of the calculation could be multiplied with the predicted price to adjust it accordingly to the change of club by the player. As it is probable that this may be an external factor that affects the price of a player without including the promise of attributes. This could potentially help us learn more information on the price the EA Sports FC market decides on a player.

We can now delve into the process of investigating the issue of bias in the transfer prices of icons in UT.

Data Collection

The data on the attributes of the various icons observed in FUT was collected from a dataset on Kaggle.com, while the data collected on the prices was obtained from futbin.com. Care was taken on the date chosen to collect the data. From experience, prices are usually high when the game starts, they then stabilise, deflation begins around the Christmas period when many children get a copy of FIFA (now known as EA Sports FC) as their Christmas present and deflation peaks towards the end of the season. The simple explanation for this trend being that new equilibrium

points on the prices are set as the supply and demand of cards changes across the game season. I picked a date a few weeks before Christmas as the ample time to collect data on the prices.

Source: Brilliant.org

This is a supply and demand curve and it can be applied to our task. In this case quantity, represents the quantity of cards in the game. As supply increases, it can be interprated as a shift in the supply curve (the red line) to the right. This results in the equilibrium price reducing by some

margin. When supply decreases, this can equivalently be represented as a shift in the equilibrium price to the left, resulting in an increase in the equilibrium price. These rules also apply to the demand curve except the effect on the equilibrium price is opposite when demand increases/decreases as opposed to the supply curve. The main way to add more players into the game, is to open more packs.

Packs contain a variety of players and can be purchased through FC Coins or FC Points. As more players get their hands on the game, more packs are opened adding more EA FC UT cards into the market. Towards the end of the game, it is common for demand to fall as players await the new edition of the game that will arrive the following season.

For the Christmas period example, we can take the fact that the price lowers after Christmas to mean that the magnitude of shift in the supply curve is much greater than the magnitude of shift in the demand curve. After all, there is a limit to how many players one can have in their team,

let alone in their squad during a matchday.

As gleaned from the study of the transfer prices of Neymar's base card, there is usually a drop in prices during the Christmas period.

The Model

Many economists and statisticians have taken on the arduous task of modelling player's transfer market valuations together with the wages players earn. Indeed, soccer is not known for its use of statistics to model the price of players and its understandable why given all of the things to consider. I looked at the literature to see whether there were methods I would be able to learn from them for my analysis on the prices of soccer player online fantasy market of FIFA 19. Tom Decroos, Lotte Bransen and others in the paper titled "Actions speak louder than words: Valuing Player's Actions in Soccer" took an interesting approach where they calculated a ranking based on how a player's actions relate to their chances of scoring. In their ranking system they write that "an action valued at +0.05 is expected to contribute 0.05 goals in favor of the team

performing the action, whereas an action valued at -0.05 is expected to yield 0.05 goals for their opponent.". Using this model they were able to quantify the performance rating of players across the board, even players that who do not shine in traditional statistics but are important to the team nonetheless. They then cross referenced the performance rating they calculated with the players transfer values on transfermarkt (online website that gives the transfer valuations for players) and achieved good results. Some predictions that they made over such as that of which player would best fill the void left by Cristiano Ronaldo in Madrid came to pass by looking at the players that could match the performance rating and action values that he exhibited .

> *Similarly, Real Madrid lost their all-time top scorer Cristiano Ronaldo in the summer of 2018. The struggling club appears in desperate need of a suitable replacement. Manchester United's Marcus Rashford and Chelsea's Eden Hazard have both been linked with moves to Madrid. However, Figure 2b shows that neither comes close to replicating Ronaldo's incredible finishing skill. Moreover, Ronaldo exhibits a higher total shot value per 90 minutes than Rashford and Hazard combined. While Hazard outperforms Rashford in every aspect, Rashford is closer to Ronaldo in terms of style as both rate similarly for passing and dribbling. If Real Madrid want to stick to their current playing style, our analysis suggests that the 21-year-old Rashford would be the better choice. However, if their aim is to immediately strengthen their team, then the 28-year- old Hazard would be the preferred choice as he is a better player regardless of his specific playing style. — Actions Speak Louder than Words by Tom Decroos, Lotte Bransen and et al*

Oliver Muller and others in the paper titled "Beyond Crowd Judgements: Data — driven estimation of market value in association football", took a more traditional approach for which they modelled the market value of a

player using statistics such as age, goals, assists, minutes played, popularity etc. They made four models, with the first one acting as a baseline model with only a constant and previous market value modelling the market value, and then incrementally increasing the number of variables per model until the fourth one, which included statistics for popularity such as the Log of Youtube Videos. As their dependent variable was a logarithm, the coefficients for their variables showed the percentage that the variable contributed to the estimate. They found that number of goals scored increased the market value by 2.6%, that the ratio of successful passes could increase a players value up to 30% .

The most important venue of transfer prices in soccer is Transfermarkt, a website that crowd sources valuations from people around the world and weighs their input hierarchically depending on how qualified the user is to give their opinion. Transfermarkt is so accurate that it is used by real players and clubs in negotiations. The accuracy of the crowdsourcing method done by Transfermarkt is very interesting given the individual player rating on FIFA are done in a similar way, where individuals volunteer from various leagues who attend games in various local leagues and may be season holders update the data on each of the players as well as
their likeness. These updates are oversaw and verified by the EA Team of Producers. Some researchers have attempted to create models that could match the transfer value estimates and achieved good results but not better than transfermarkt. Further review of the literature also showed that there were other researchers who created hedonic models for wages of soccer players based on player attributes given on *sofifa.com*. These researchers also achieved great results and particularly found that passing ability proved to be very significant in the determination of

players' wages. This gave me confidence that a similar feat could be performed for transfer prices in FUT and the following was my model, which was of semi-log functional form.

The physical attributes I looked at were, jumping ability, pace, strength as well as heading ability. The technical attributes I looked at were dribbling, dexterity (a player's ability to use their weak foot), skill rating (the level of skill a player can perform in game) and the passing ability as rated by EA Sports. There were also three dummy variables: two representing whether a player was a defender or midfielder and one checking if a player possessed the finesse shot trait. The reason there was no dummy variable for being a forward was to avoid the dummy variable trap which could introduce perfect multicollinearity. In addition, out of the three categories of being a player: defender, midfielder and forward, I assumed that the forward dummy variable would provide the least information in the model. Reason being that I believed it would introduce redundancy as these attributes would be expected from an expensive

forward, and so I believed they would provide more information on defenders and midfielders. The other dummy variable was for the finesse shot trait which allows players to score by kicking the ball around a goal keeper. As for the other attributes, my choice for many of these attributes came about as the result of careful study the most expensive icon card in each position, and comparing them to other icons in similar positions trying to see which strong attributes they had that were different from the rest of the players in similar positions.

"Let's not dismiss their business plan just because the font makes it hard to read" — Daniel Kahneman

Fifa has added a plethora of traits(now known as play styles) for different roles in the game. For strikers, some of these include *deadball* which increases the quality of set pieces/distribution you can have with a player*, finesse shot* which has been mentioned before and *powershot* amongst others. Surprisingly they have classified the *trivela,* which is a technique some players have that allows them to hit a shot from the outside of their foot as a physical playstyle. Closer inspection would reveal that this trait

is not much different from the finesse shot empirically (to score with a finesse shot one must hit the ball from the inside of the foot), the implication would be that perhaps this shot has a more diverse set of uses across the pitch. Another interesting thing is that instead of classifying being a *poacher* as a playstyle, it has been included among EA Sports arsenal of tactical manoeuvres in FC IQ — their tactical system engine. This is surprising, as it would imply that all players can play the *poacher* role. In reality, very few players in real life can be classified as true poachers.

"Some people have a knack for being in the right place at the right time. They just happen to be under the tree when the apple plops into their lap. Some people have to bash and butt at the base of the tree for an awfully long time until the exhausted apple stalk can bear the weight no longer. I am definitely in the second category" — Boris Johnson, Unleashed

One of the most impressive goal poachers to grace the Premier League could only sustain his brilliance for one season and what a season it was for him. He terrorised defences across the Premier League, both big and small teams were defenceless from his movements.

For more information on what the attributes entail, please visit https://www.fifauteam.com/fifa-18-attributes-guide/

Quick Snapshot of the Data obtained from **Kaggle Datasets and FUTBIN**

Results

Regression powered by Stata ©

After running the regression, I obtained an R-squared statistic of 0.7250 for my model which was quite decent. When taking a look at the variables we see that Strength had the highest amount of significance when looking at the t-statistic followed by pace and heading respectively. The physical attributes each make up the variables with the highest t-statistics. Surprisingly, variables such as passing and dribbling were not significant at the 5% percent level. We also see that the dummy variables of being a midfielder or a defender were also significant at a 5 % level. The same cannot be said for the finesse shot trait which is also surprising.

Limitations to Methodology — Probability with a hint of Number Theory

Upon taking a second look at the results, we see that out of the 11 attributes selected to detect bias in the transfer prices, 3 of them were averages. Pace is an average of sprint speed and acceleration. Passing is an average of vision, crossing, free kick accuracy, short pass, long pass

and curve. Dribbling is an average of agility, balance, reactions, ball control, dribbling, and composure.

“In probability theory, the central limit theorem (CLT) states that, under appropriate conditions, the distribution of a normalised version of the sample mean converges to a standard normal distribution” — (https://en.wikipedia.org/wiki/Central_limit_theorem)

Source: Wikipedia (https://en.wikipedia.org/wiki/Central_limit_theorem)

That is to say that as the size of the sample grows, the average of that sample approaches the expected value. This would give the impression that the variables of passing and dribbling should be giving more accurate data on the players than standalone variables such as strength, jumping and heading. It would also mean that these variables are less prone to extremities as some variables would make up for discrepancies in others. In passing for example, vision would make up for a discrepancy in the short pass or curve would make up for a discrepancy in the long pass.

There is, however, another side to this, the subset of players I chose to look at were icons. There is a very good chance that all of them have great

fundamentals, i.e that they are all good on the ball. Therefore, there may not be a very good way to distinguish an icon who is very good at passing the ball to one who was good at passing the ball.

Passing Statistics of Beckham's medium card in FIFA 23 (EA Sports)

Passing Statistics of Bergkamp's medium card in FIFA 23 (EA Sports)

The difference in 9 passing points may not be enough to highlight how good Beckham was at passing the ball and Bergkamp himself was not a shabby passer of the ball. Perhaps exponentiating the passing and dribbling values before running the regression would provide greater information on the prices. As for pace, EA Sports is extremely strict with how they grade the pace of players, it is probably one of the most scrutinized statistics in the game. It may not need to be exponentiated.

PS: Upon further consideration, simple exponentiation of the passing and dribbling values may not be enough. The following two players may provide a reason why.

Luka Modrić's passing stats in EA FC 25

Joshua Kimmich's passing stats in EA FC 25

Lets say we were exponentiating all passing values to the third value, the value we would have for passing would be 681472 for both Kimmich and Modrić. While this value may show that there is a big difference from a player with a passing score of 87 such as Antoine Griezmann, whose new value for passing would be 658503. It may not tell the full story.

Proof by Contradiction (through Linear Algebra)

> *"All animals are equal, but some animals are more equal than others" — George Orwell , Animal Farm*

Assumption: The overall passing statistic for players in EA FC UT is accurate and linear.

This would mean that there exists some linear combination of the attributes that make up that the passing statistic that satisfies the overall

passing rating. This would also inherently prove that the passing statistic is **linear** in the sense that it accurately judges the value of players and the scale can be trusted. And that there is mathematically sound way of giving one player a higher rating than the other through the attributes they have provided to quantify a player's passing rating. (It would be very disappointing to know that after all the crowdsourcing judgements of the attributes that make up the passing statistic, that the end result displayed by EA Sports is ultimately arbitrary)

A simple test would be to try to solve a system of six linear equations that each have the same result.

The group of players starting from Bruno Fernandes going down to Parejo each have the same passing rating.

We can use the individual values of the attributes that make up the passing rating of each of those players to try and see if we can come up with some reasonable coefficents. In this case we expect all of the individual attributes of vision, crossing, free kick accuracy, short pass, long pass and curve to contribute positively to the score. Why select players who all have the same passing rating ? The reason is that we can

set the coefficients to be **axioms** as something that experts can widely agree on as a basis for future and more complicated mathematical understanding or rigour.

> *An **axiom**, **postulate**, or **assumption** is a statement that is taken to be true, to serve as a premise or starting point for further reasoning and arguments. The word comes from the Ancient Greek word ἀξίωμα (*axíōma*), meaning 'that which is thought worthy or fit' or 'that which commends itself as evident'. — Wikipedia, Axiom, https://en.wikipedia.org/wiki/Axiom*

The building blocks of mathematics are axioms, in essence these are statements that cannot be proved to be correct but can be widely be assumed/accepted to be correct. From these statements is where all great/small mathematical proofs arise from. By forming coefficients from these set of players, we can have a starting point on what the they truly present from the perspective of EA FC and use these to judge the other ratings they have provided, allowing us to see whether or not they are linear or accurate in any shape or form.

Kimmich, Bruno Fernandes, Grimaldo and Modrić's passing rating

Hansen and Parejo's passing rating

When we input these values into a system of linear equations we get the following result:

Coefficients of the attributes of the passing rating

U corresponds to **Vision**, **V** corresponds to **Crossing**, **W** corresponds to **Free Kick Accuracy, X** corresponds to **Short Passing**, **Y** corresponds to **Long Passing** and **Z** corresponds to **Curve.**

Contradiction: To show that the passing statistic in EA FC is not linear, let us apply these coefficients to a different player such as Antoine Griezmann Rio Ferdinand or David Beckham, who each have passing scores of 87, 65 and 92 respectively. What does the equation ultimately give us.

Antoine Griezmann’s passing attribute scores in EA FC 25

His Predicted Passing Rating — Credit to Wolfram Alpha

Rio Ferdinand’s passing attribute scores in EA FC 25

His Predicted Passing Rating — Credit to Wolfram Alpha

David Beckham's passing attribute scores in EA FC 25

His Predicted Passing Rating — Credit to Wolfram Alpha

The equation gives us a rounded prediction of 86 as a passing rating for Griezmann which is clearly off the mark by a margin of 1. For Rio Ferdinand, the equation gives us a rounded prediction of 60 which is off the mark by a margin of 5. Surprisingly for David Beckham, the equation undershoots by a margin of 1. Meaning that these coefficients cannot be accurately extrapolated to the rest of the game. Since these overall ratings were used to determine the price of players, it would make sense why they were insignificant in the bigger picture of prices. In a sense it does appear that after getting the ratings for the individual attributes, the engineers pick arbitrary scores for the overall passing rating. This may be to appease players as many players do use these scores for bragging

rights in their changing rooms in the period before the season begins. Maybe in the case of Griezmann, it may be a way to hide that his actual passing rating decreased from the previous year as his passing score in EA FC 24 was also 87 so that he keeps buying and promoting the game. This of course, is under the assumption that these coefficients accurately represent the importance of each attribute to the passing ability of a player. In this case **Vision** and **Crossing** are seen to be the most salient factors and not the **Short Pass** or the **Long Pass** while **Free Kick Accuracy** and **Curve** are very insignificant in comparison in what they contribute to the overall passing rating.

QED

This would mean that more complex math may be needed to understand the implication of EA FC' passing ratings and that matching the individual attribute values to the overall rating is an optimization problem with linear inequality constraints. Linear programming or Quadratic programming may be needed to make sense of these ratings. That in itself would mean that the system of equations are inherently nonlinear which means that they provide no information on who may be a better passer than the other. This is pretty clear when you look at the passing rating of Rio Ferdinand.

Why should they be linear ?

The attribute values that make up the passing index value are different for both Kimmich and Modrić. In reality, vision could be more valuable than free kick accuracy or a short pass could be more valuable than a long pass. That is to say that given these attributes, there could be a numerical difference between their resulting overall passing rating. What may be needed may be a different weighted interpretation of the passing values, which can be ultimately exponentiated if need be so that there is a clear linear distinction of the passing ability of different players.

Example of the new passing index that could be used for the regression model

What makes this interesting ?

This is interesting as most passing measures can be quantified into simple statistics such as number of successful/unsuccessful passes, number of assists, number of through balls, number of dead-balls, number of crosses etc. But what about the curve or vision, players could be subject to a few training drills by EA FC to get a better picture of their curving ability or vision before each season, if they want a verified curve or vision rating.

How would I approach this ?

To have this different approach to get the weighted attributes that make up the passing index, I would use classical trickle down economics.

It is clear that EA FC's method of grassroots economics of using tautologies and axioms to build up an accurate passing rating is fulled of fulls and can even be considered to be shameful by some.

By trickle down economics, I mean I would distribute all my energies to create a passing index based on the results of assists, through balls, accurate corners, short pass completion, long pass completion... From this point onward I would then use these indexes to create a passing rating for each player. I would then use these values to create coefficients for the passing attribute values that have been crowdsourced. I would then ultimately use these values to create an EA FC passing index that would hopefully be significant in determining the actual prices of players.

Example of a formula for my target values for passing

Carrying out something like this would provide more insight on the impact of passing on the transfer price of a player in EA FC UT.

Inspiration

Part of this novel approach was inspired by the proof for the 3 state Busy Beaver problem by Shen Lin and Tibor Rado. (https://dl.acm.org/doi/pdf/10.1145/321264.321270).

- The proof for BB(3) was done by Shen Lin and Tibor Rado in a paper called Computer Studies of Turing Machine Problems in 1965
- It involved three major steps: Reducing the search space, discarding the turing machines that run less than or equal to the conjectured

number of times, deciding that the rest halt by finding recurrence patterns that occur when they are running

- There are about 17 million possible 3 state Turing Machines, at the time Lin and Rado did this, this would have been computationally expensive to do on a computer.
- They were able to do this through normalization of the machines.
- This reduced the search space to 82, 944 machines which they separated into four equal lots
- It had been conjectured that the BB(3) = 21, so the approach Lin and Rado took was running an algorithm that discarded machines that ran in less than or equal to 21 steps while recording the maximum number of ones each of those machines wrote on the tape.
- As for the machines that did not halt, they were analyzed for any patterns that may signal that these machines did not halt.
- Total recurrence and partial recurrence.
- The main idea behind these patterns was that if a machine arrives at a state and scanned symbol combination it has encountered before, holding tape conditions to be constant, then the machine would never halt.
- This approach discarded all but 40 holdout machines which upon further analysis by hand it was determined that they did not halt.

A note on extrapolation and linearity from ChatGPT o1

Extrapolation involves predicting outcomes outside the range of observed or known values. When dealing with nonlinear systems, the process of extrapolation becomes more complex and riskier compared to linear cases. Here's why:

1. ***Complex Behavior Beyond Known Bounds:***
   In a linear system, the relationship between variables and the outcome is a straight line (or a flat hyperplane in higher dimensions). While extrapolation can still be inaccurate if extended too far, the shape is simple and predictable. In nonlinear systems, the relationship can bend, curve, or even change direction in ways that aren't easily anticipated. This means small steps outside the known data range can lead to large and unexpected changes in the predicted outcome.

2. ***Multiple Possible Outcomes for Similar Inputs:***
   Nonlinear functions can exhibit complex behaviors such as local maxima, local minima, plateaus, or oscillations. When you move beyond the known data range, you may enter a regime where the function behaves very differently. For example, a function might have been slowly curving upward within the known range, but just beyond that range, it could sharply drop. Such shifts are not predictable without a detailed understanding of the underlying nonlinear structure.

3. ***Increased Sensitivity to Uncertainty:***
   Nonlinear systems are often more sensitive to small variations in input values or parameter estimates. When you extrapolate, you compound uncertainties — small errors in parameter estimation or assumptions can translate into disproportionately large errors in predictions. This sensitivity makes trusting extrapolated results from nonlinear models more challenging.

4. ***Greater Difficulty in Model Validation:***
   Validating extrapolated predictions is difficult since, by definition, you have no observed data in the range to which you are extrapolating. In a nonlinear context, the lack of straightforward structure means you have fewer reliable cues to guess how well your model might behave outside the observed domain.

5. ***Necessity of Domain Knowledge:***
   Because nonlinear systems can behave in unexpected ways, relying purely on the mathematical model is risky. Domain expertise becomes even more critical: understanding the underlying physical, biological, economic, or other contextual factors can guide reasonable extrapolations and help spot implausible forecasts.

In Summary:
Nonlinear systems make extrapolation less predictable, more sensitive to errors, and more dependent on domain knowledge. While linear systems at least offer a straight-line guess when projecting beyond known data, nonlinear systems can produce a wide range of behaviors, making extrapolated predictions significantly less reliable without careful consideration and additional constraints.

Further thoughts

As this research was made based on FIFA 19, a further study can be made on EA FC 25 to see whether much has changed. Since then a tactical game engine has been added, and it is possible the game mechanics have changed. The most expensive icon card is still Ronaldo, O Fenomeno.

Current Study of Today's Top Young Talent

Most expensive base card in EA FC 25

Kylian Mbappé, the Frenchman, is one of the highest rated players in the world. He started receiving world acclaim due to his startling performances for AS Monaco in the 2016–17 season of the Champions League. He became world renowned soon after due to his performances during France's triumph in the 2018 FIFA World Cup where he became

the second teenager to score in the cup's final after the great Pelé. As a tribute to the great young man is the following poem by one of his fellow countrymen, Eric Cantona, where he insidiously describes the state of the world for footballers and non-footballers alike. It does also happen to remind me of a poem by Wilfred Owen.

Current Ballon d'Or Holder

The Makélélé role

"*In fact, the simple term 'defensive midfielder' didn't do Makélélé's role justice; he was described as a 'holding midfielder', a 'sitting midfielder', a 'screening player', all of which implied his role was based around positioning and protecting rather than simply tackling. And then, eventually, it just became the 'Makélélé role' — position in itself — an extremely rare honour. The surprising aspect of Makélélé's position, however, is that he started playing the Makélélé role — the sole holding midfielder at the base of a trio or a diamond — very late in his career. Far from it being his natural position, Makélélé essentially dropped deeper and sacrificed himself for the team in the autumn of his career, precisely what Vieira and Keane were unable to do*" — Excerpt from *The Mixer* by Michael Cox

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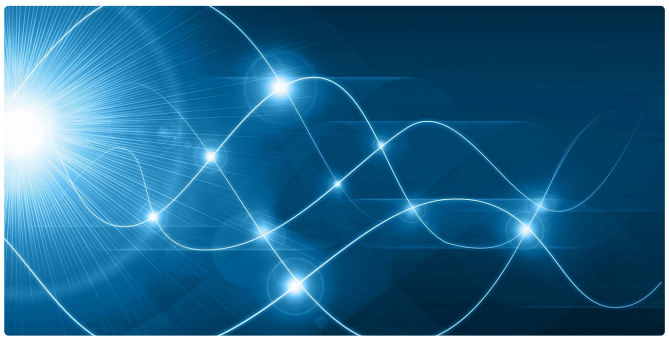


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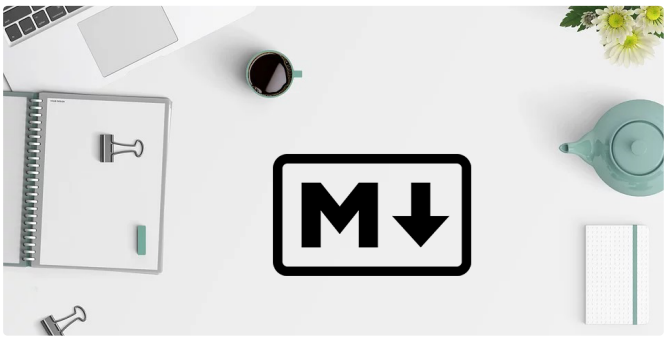


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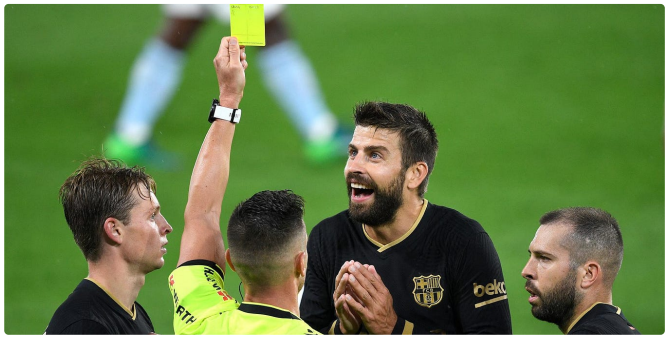
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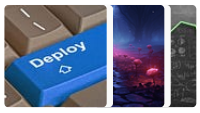
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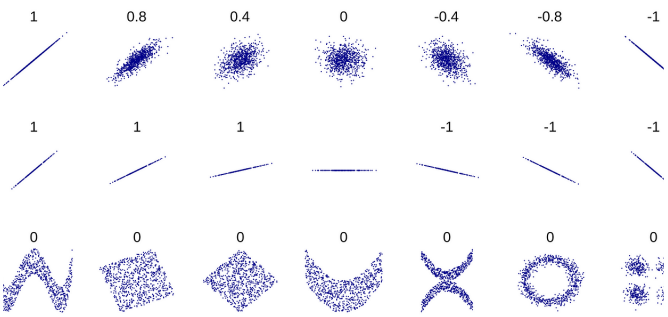


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